

Handwritten Digit Classification Using Convolutional Neural Networks

Automated Visual Feature Extraction & High-Precision Digit Recognition on the MNIST

Benchmark Dataset

SUBMITTED BY

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Introduction

- ✓ Handwritten digit recognition is a foundational computer vision benchmark in modern deep learning research.
- ✓ CNNs automatically extract spatial feature hierarchies directly from raw image pixels without manual filter engineering.
- ✓ The system classifies single handwritten digits (0-9) utilizing 70,000 standardized grayscale images.
- ✓ Real-world applications span bank check processing, automated postal sorting, document OCR, and form parsing.



Key Objectives



CNN Fundamentals

Understand convolutional layers, pooling, activation functions, and feature extraction mechanisms.



MNIST Exploration

Load, analyze, and inspect pixel distributions across 70,000 digit images.



Data Normalization

Preprocess raw 8-bit integer pixel values (0-255) into scaled floating point ranges (0.0-1.0).



Build CNN Model

Construct a multi-layer sequential deep CNN model using Keras and TensorFlow APIs.



Train & Evaluate

Train over multiple epochs, track categorical cross-entropy loss, and optimize validation accuracy.



Predict Accurately

Perform inference on unseen test samples with confidence score metrics.

Dataset Overview

70,000

Total Images

60,000

Training Set

10,000

Testing Set

28x28

Image Resolution

10

Output Classes (0–9)



Grayscale Pixel Matrix

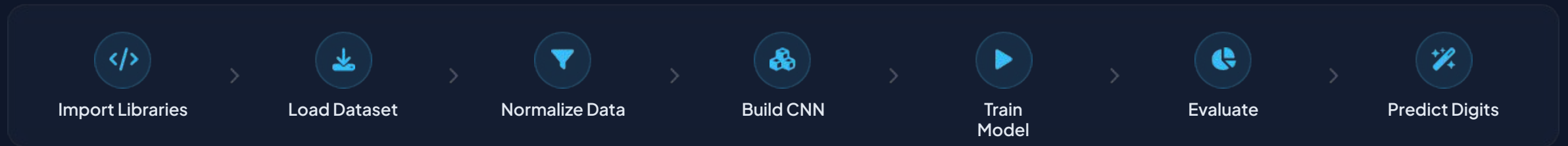
Each image consists of a 28×28 bounding grid totaling 784 features. Values represent grayscale brightness intensities where 0 is background and 255 is full stroke intensity.



Balanced Class Distribution

The dataset exhibits an approximately uniform distribution across digits 0 through 9, eliminating class imbalance bias during neural network weight updates.

Methodology & Preprocessing



TECHNOLOGIES & TOOLS UTILIZED

 Python 3.x

 TensorFlow

 Keras API

 NumPy

 Matplotlib

 Google Colab

CNN Architecture Design

Layer (Type)	Output Shape	Param #
conv2d (Conv2D 32, 3x3)	(None, 26, 26, 32)	320
max_pooling2d (2x2)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D 64, 3x3)	(None, 11, 11, 64)	18,496
max_pooling2d_1 (2x2)	(None, 5, 5, 64)	0
flatten (Flatten)	(None, 1600)	0
dense (Dense ReLU)	(None, 128)	204,928
dropout (Dropout 0.2)	(None, 128)	0
dense_1 (Softmax)	(None, 10)	1,290

TOTAL PARAMETERS
225,034 (879 KB)

OPTIMIZER & LOSS
Adam | Categorical Crossentropy

HYPERPARAMETERS
Epochs: 5 | Batch Size: 128

Model Training Process

Training vs Validation Accuracy



Training vs Validation Loss



Results & Evaluation

TEST ACCURACY

99.1%

✔ High Precision Classification

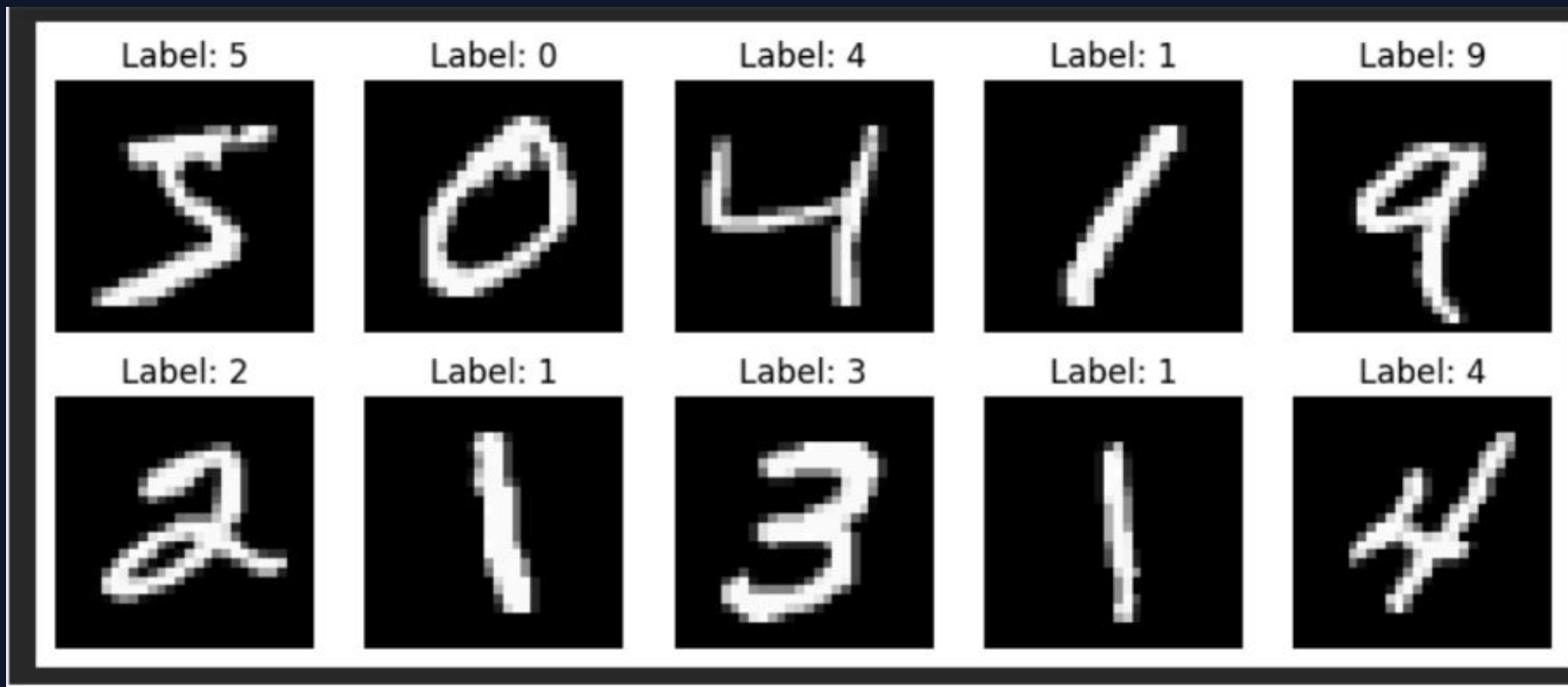
TEST LOSS

0.032

Minimal Cross-Entropy Error

Digit Class	Precision	Recall	F1-Score	Support
Digit 0	0.99	1.00	0.99	980
Digit 1	0.99	1.00	1.00	1135
Digit 2	1.00	0.99	0.99	1032
Digit 3	0.99	1.00	0.99	1010
Digit 4	0.99	0.99	0.99	982
Digit 5	0.99	0.98	0.99	892
Overall Accuracy	99.1% Benchmark Accuracy	10,000		

Sample Predictions



Project Conclusion



High Performance

Successfully designed and trained a 2D CNN achieving 99.1% test accuracy and 0.032 loss on 10,000 unseen test digits.



Automatic Feature Learning

Demonstrated the capability of CNN convolution filters to learn hierarchical spatial edges and curves directly from pixel data.



Effective Regularization

Incorporating Dropout (0.2) and Batch Normalization prevented overfitting, maintaining minimal validation loss gap.



Industry Relevance

Provides a scalable blueprint for modern OCR automation across banking, logistics, and document parsing workflows.

Future Scope & Q&A

🔗 Future Extensions

- Data augmentation (rotation, scaling, noise injection) to boost robustness.
- Deployment via Streamlit / Flask for real-time interactive canvas drawing.
- Extension to EMNIST for combined alphanumeric character recognition.



Thank You!

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